

CORRELATION BETWEEN ROOT DENSITY AND SOIL MOISTURE OF CARAGANA MICROPHYLLA IN XILINHOT GRASSLAND

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ABSTRACT

Soil moisture is generally considered to be a major limiting factor for root growth of shrubs in the arid and semi-arid regions. However, as a typical shrub species, the correlation between root distribution and soil moisture of *Caragana microphylla* has not been well investigated. Ground-penetrating radar (GPR) has the advantage of acquiring locations of roots quickly and simply, therefore, we used GPR to acquire locations of roots in the quadrat of 30m*30m, and soil moisture data was acquired by soil drilling method. After processing data, we analyzed the correlation between root density and soil moisture in vertical and horizontal directions. Following conclusions were obtained through vertical and horizontal analysis: In the vertical direction, root density was positively correlated with soil moisture, but the soil moisture was lower in the dense absorbing roots' distribution region, even the preferential flow could enrich soil moisture. Root density and soil moisture also had a positive correlation in horizontal direction, but they were inversely related in the region of dense absorbing roots.

Index Terms—Ground-penetrating radar, soil moisture, *Caragana microphylla*, root system

1. INTRODUCTION

In arid and semi-arid regions, water is generally considered to be a major limiting factor for root growth of perennial shrub growth [1]. The water use strategy can largely decide whether they can survive through drought and precipitation period [2]. Exploring the correlation between shrub roots distribution and soil moisture in arid and semi-arid regions is significant to study shrub growth, which is also helpful to study the underground root system growth pattern of perennial shrubs. In previous studies, some researchers have explored the relationship between root distribution and soil moisture of *Caragana microphylla* in the vertical direction

using excavation method [7]. However, due to limitations of this method, they only studied the relationship between the number of roots and soil moisture in the vertical direction, which failed to explain the relationship horizontally. In recent years, studies have shown that the preferential flow formed by the precipitation of shrubs was stronger than the preferential flow of grass, which means the shrub has larger soil moisture in deeper position under the action of the preferential flow [3-5]. Researchers have also reported the age and root function of shrubs. Li et al. found that the density of *Caragana microphylla*'s shallow roots significantly increased with the increasing age, making it possible to use precipitation effectively [6]. Alamusa et al. confirmed that the absorbing roots of shrubs had a critical impact on soil moisture, and the soil moisture in lower layer of the absorbing roots distribution area was much lower [7].

The objective of this study was to analyze the correlation between root density and soil moisture with GPR data and soil moisture data. GPR has the advantage of acquiring the location of roots quickly and easily, which convenient for us to analyze in both vertical and horizontal directions.

2. DATA AND METHODS

2.1. Study area

Inner Mongolia's Xilingol League is the most typical grassland distribution region in northern China. It is dominated by plateaus and plains in average elevation of 1000 meters. The location selected for this study is Xilinhote, which is the capital of Xilingol League. It belongs to the mid-temperate semi-arid continental climate, and is controlled by high-pressure airflow in Mongolia during the cold dry winter. The annual average temperature and precipitation are respectively 0-3 degrees Celsius and 350mm while most precipitation occurs from June to September. The study area was selected as a quadrat of 30m*30m in Xilinhote, including 48 *Caragana microphylla* and other non-shrub vegetation.

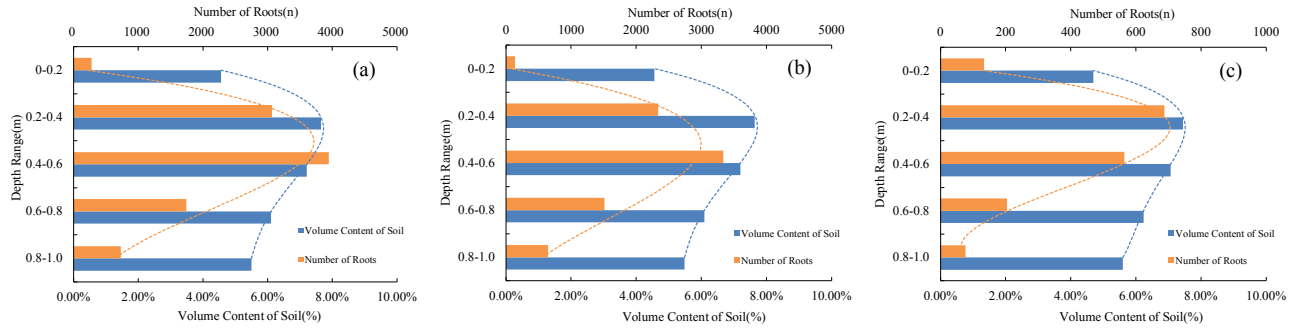


Figure 1 Distribution of roots and soil moisture at different depths in the whole study area (a), outside the shrubs' growth area (b) and within the growth area of shrubs (c).

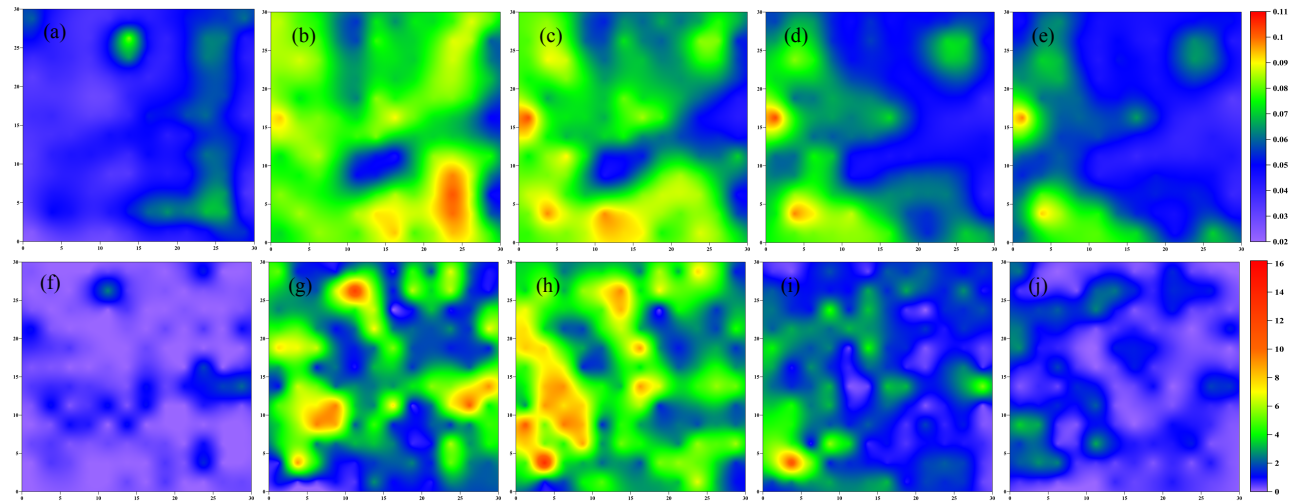


Figure 2 Distribution of root density and soil moisture at different depths in the whole study area. Figures a to e represent the distribution of soil moisture at different depths, the warmer the color, the larger the soil moisture, and the depth ranges are 0cm-20cm, 20cm-40cm, 40cm-60cm, 60cm-80cm and 80cm-100cm, respectively. Figures f to j represent the distribution of root density at different depths, the depth and color meanings are consistent with Figures a to e.

2.2. Data

GPR data and soil auger data were acquired from the study area on August 17, 2018. There was no precipitation in or around the study area within two weeks prior to data acquisition. Starting at the point with coordinates (1.5, 1.5), we took a set of soil auger data every 3m in the direction of X and Y axes respectively, a total of 100 sets of soil auger data were acquired, and each set of soil auger data was divided into 5 different layers according to soil depth (0-20cm, 20-40cm, 40-60cm, 60-80cm, 80-100cm). Spatial coordinates of roots were acquired by GPR, which is fast and efficient. GPR measured data were acquired every 25cm in the positive direction of the X coordinate axis, as was the Y direction, and a total of 240 measurement lines were acquired. Soil moisture data and root position data were re-acquired through the data processing in laboratory. The ground growth parameters of the *Caragana microphylla* in this study area were measured at the same time, including the long diameter, short diameter, and growth center.

2.3. Methods

The stratified method was selected to analyze the relationship between root density and soil moisture in different layers of underground study area. According to soil depth, 5 layers were divided from soil surface and each layer was 20cm. For each layer in horizontal direction, large quadrat of 30m*30m was divided into small quadrats of different sizes to analyze the relationship using least squares regression method. In order to acquire the value of soil moisture on small quadrats, original soil moisture data was resampled with the method of kriging interpolation, and then calculated the average soil moisture of the effective area as the soil moisture value of small quadrats. Root density of each small quadrat was acquired by calculating the number of root points per unit area. In vertical direction, root number and soil moisture were acquired with statistical method to compare the distribution of roots and soil moisture at different depths.

After global analysis, the large quadrat was divided into two parts to local analysis including inside and outside shrubs

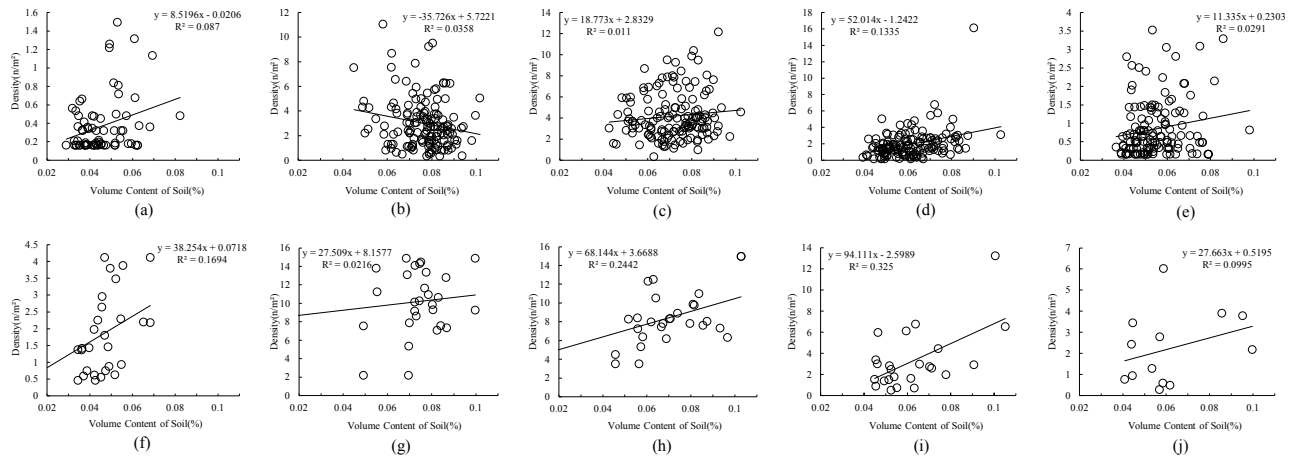


Figure 3 The correlation between soil moisture and root density using least squares regression on the growth area of *Caragana microphylla* and outside the growth area. Figures a to e represent the correlation of outside area at different depths, and the depth ranges are 0cm-20cm, 20cm-40cm, 40cm-60cm, 60cm-80cm and 80cm-100cm, respectively. Figures f to j represent the correlation of the growth area of *Caragana microphylla* at different depths, depths are consistent with Figures a to e.

respectively. We considered that growth locations of shrubs may have an effect on root distribution, so the whole study area was divided into two separate areas for study, including the growth area of *Caragana microphylla* and outside the growth area. The growth center of *Caragana microphylla* as the center of the circle, and the long diameter of *Caragana microphylla* as the diameter, then two local study areas were acquired. Since the quadrat was divided according to circular growing areas, the effective area of small quadrats and the effective root number of small quadrats were considered in the calculation.

Table 1 Correlation between root density and soil moisture inside and outside the area covered by *Caragana microphylla*. *Significant at $P < 0.05$ and **Significant at $P < 0.01$.

Soil Depth (cm)	Pearson Correlation	
	Outside	Inside
0-20	0.191*	0.351
20-40	-0.184*	0.072
40-60	0.105	0.384
60-80	0.365**	0.561**
80-100	0.170*	0.448*

3. RESULT AND ANALYSIS

3.1. Vertical direction

In vertical direction, statistical method was used to analyze the relationship between soil moisture and root density in local and global scale. The distribution patterns of soil moisture and roots were similar in the whole study area, (Fig. 1a) showing a tendency to rise from the surface and then decrease. The maximum of soil moisture appeared in the depth of 20 cm to 40 cm, but maximum root density appeared in a deeper layer, which was consistent with findings by

Alamusa et al. In local analysis, distribution trends of roots and soil moisture outside shrubs were consistent with the global analysis. (Fig.1b) There were differences in the distribution pattern within the growth area of shrubs, roots' distribution was highly consistent with the soil moisture's distribution, which peaked in the depth of 20 cm to 40 cm. (Fig.1c)

Although the preferential flow caused by precipitation could supplements the soil moisture around roots, the outside shrubs' growth area with a dense absorption root could absorb more water from a depth of 40 cm to 60 cm. For the *Caragana microphylla* growth area, preferential flow brought more water than roots can absorb, which meant more roots, more soil moisture.

3.2. Horizontal direction

In global analysis, the large quadrat of 30m*30m was divided into small quadrats of 2.5m*2.5m. Root density and soil moisture of each small quadrat were calculated at different depths, and plotted as color level maps. (Fig.2) At depths of 0-20cm, 60-80cm and 80-100cm, distribution of soil moisture and root density was similar, the more roots there were, the more soil moisture there was. At the depth of 40-60cm, this positive correlation was not so obvious, but it still showed a characteristic of high root density with high soil moisture. At the depth of 20-40cm, the correlation between root density and soil moisture showed an opposite trend, that was, the soil moisture in the place where the root density was large was relatively low.

Scatter plots of local analysis are shown in Fig. 3. In the depth between 20cm-40cm without the area of *Caragana microphylla* covering, root density had an inverse correlation with soil moisture, that was, the more roots, the less soil moisture. This opposite correlation between root density and soil moisture may be caused by the action of absorbing roots.

Among the other nine correlations, as the roots of *Caragana microphylla* increased the soil pores, a preferential flow occurred during precipitation, which caused soil moisture enriched by precipitation around the coarse roots, showing a positive correlation between root density and soil moisture. Pearson correlation coefficient and significance are shown in Table 1.

At the depth of 60cm-80cm, the correlation between root density and soil moisture was significantly higher than other depths, showing a positive correlation. It may contain the combined effects of precipitation and groundwater. On the one hand, soil moisture was enriched by precipitation around the coarse roots, on the other hand, adsorbing roots had the property of raising groundwater up. These two reasons caused the soil moisture enriched near the roots, which shows a significant positive correlation between root density and soil moisture.

4. CONCLUSION

The whole study area was divided into 5 layers to analyze the correlation between root density and soil moisture in vertical and horizontal directions. Growth locations of shrubs were considered may have an effect on root distribution, therefore, the whole study area was divided into two separate areas after dividing into 5 layers, including the growth area of *Caragana microphylla* and outside the growth area. In vertical direction, the distribution patterns of soil moisture and roots were similar in the whole study area, showing a tendency to rise from the surface and then decrease. Due to dense absorbing roots outside the shrub growth area, the maximum of soil moisture appeared in the depth of 20 cm to 40 cm, and maximum root density appeared in a deeper layer. In horizontal direction, due to the influence of preferential flow, root density was positively correlated with soil moisture, but

it was inversely related in the area of dense absorbing roots, such as the depth between 20cm-40cm without the area of *Caragana microphylla* covering. Groundwater also had an effect, which could increase the soil moisture in dense root areas.

We did not consider the effect of the quadrats' size on the least squares method in this study. Similarly, the growth area size of *Caragana microphylla* also had an effect, which requires further research to explore. Whether the upper roots have an absorption effect on the lower soil moisture is also worthy of concern and needs further discussion.

5. REFERENCES

- [1] Breshears, D., McDowell, N., and Dayem, K., "Foliar absorption of intercepted rainfall improves woody plant water status most during drought," *Ecology*, 89(1), pp.41–47, 2008.
- [2] Gong, R., Xu, X., Tian, X., Jiang, H., Li, X., and Guan, M., "Hydraulic architecture characteristics and drought adaption strategies for three *Caragana* genus Species," *Acta Ecologica Sinica*, 38(14), pp. 4984-4993, 2018.
- [3] Li, L., Zheng, X., Li, X., and Zhang, S., "Influences of Soil Infiltration and Preferential Flow by Shrub *Caragana microphylla*," *Journal of Soil and Water Conservation*, 29(2), pp.55-59, 2015.
- [4] Beven, K., Germann, P., "Macropores and water flow in Soil," *Water Resources Research*, 18(5), pp.1311-1325, 1982.
- [5] Fury, M., Jury, W., and Leuenberger, J., "Susceptibility of soils to preferential flow of water: Afieldstudy," *Water Resources Resesch*, 30(7), pp.1945-1954, 1994.
- [6] Fu, Q., Liu, X., Liu, T., Duan, L., Wang, G., Cao, W., and Huang, T., "Water use strategy of *Caragana microphylla* during the transition from drought to rainfall," *Acta Prataculturae Sinica*, 27(8), pp.67-77, 2018.
- [7] Alamosa, Jiang, D., and Yan, T., "Correlation Between Root System Distribution and Soil Moisture Of Artificial *Caragana microphylla* Vegetation in Sandy Land," *Journal of Soil and Water Conservation*, 17(3), pp.78-81, 2003.